

RAI-Audit UK v1 - Loan Approval Governance Audit Report

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Dataset: loan_approval_dataset_A.csv

Target: loan_status (0=Approved/Good, 1=Rejected/Bad)

Protected / proxy feature: education (proxy=True)

Feature set: income_annum, loan_amount, loan_term, cibil_score, residential_assets_value, commercial_assets_value, luxury_assets_value, bank_asset_value

Splits: train=2988, validation=640, test=641

Fairness audit note: The favourable outcome is class 0, meaning Approved/Good. For Dataset A education-based proxy auditing, Not Graduate is treated as the unprivileged audit group and Graduate as the privileged audit group. SPD is calculated as the approval rate of the unprivileged group minus the approval rate of the privileged group. DI is calculated as the approval rate of the unprivileged group divided by the approval rate of the privileged group. EOD is calculated as the favourable-class true positive rate of the unprivileged group minus the favourable-class true positive rate of the privileged group. These metrics are fairness audit evidence only and do not prove legal discrimination or certify legal compliance.

GRSE interpretation note: The Governance Ready Scoring Engine is a risk score. A lower GRSE score indicates stronger governance readiness under the tested audit conditions.

Model Card and Audit - Logistic Regression

Performance Score: 90.19
Governance Ready Scoring Engine (GRSE) Score: 28.17 (lower = stronger governance readiness)
GAR Classification: Green
Deployment Decision: Low Risk - Deployable

Predictive Performance

Metric	Validation	Test
Accuracy	0.933	0.913
Precision	0.913	0.912
Recall	0.909	0.851
F1	0.911	0.880
ROC-AUC	0.978	0.967

Confusion Matrix



Governance Evidence

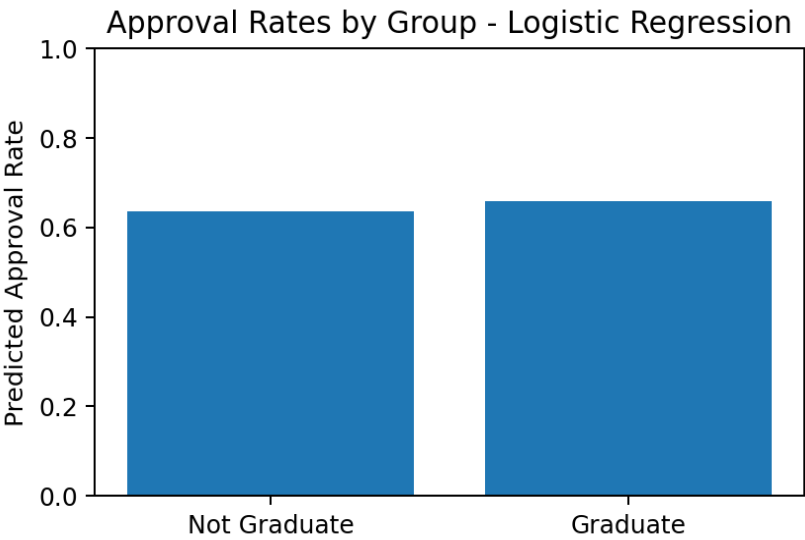
Metric	Value
Approval rate (Not Graduate)	0.635
Approval rate (Graduate)	0.659
SPD: unprivileged - privileged	-0.023
DI: unprivileged / privileged	0.965
EOD: unprivileged - privileged	-0.024
Fairness risk	0.122
Flip-rate (noise)	0.014
Flip-rate (missingness)	0.016
Robustness risk	0.148

Metric	Value
Explainability risk	0.397
Top feature dominance	0.479

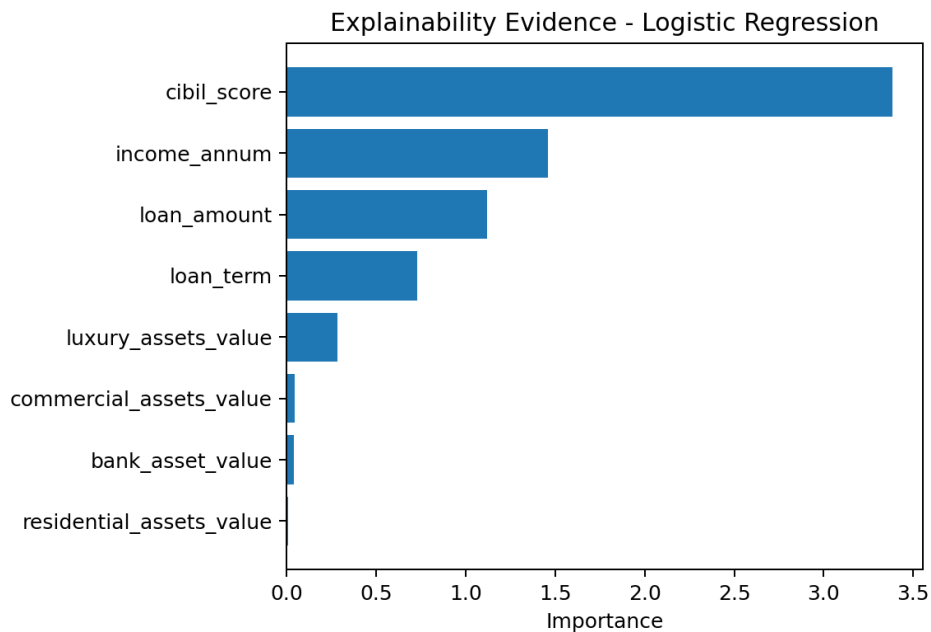
Governance Ready Scoring Engine (GRSE: lower score = stronger governance readiness)

GRSE Component	Risk Score Contribution
Fairness component	4.26
Robustness component	2.96
Explainability component	9.94
Governance baseline component	11.00
Total GRSE score (lower = stronger governance readiness)	28.17

Fairness Evidence



Explainability Evidence



Top Features

Feature	Importance
cibil_score	3.383981
income_annum	1.458905
loan_amount	1.119585
loan_term	0.726603
luxury_assets_value	0.285557
commercial_assets_value	0.043677

Governance Mitigation Factors and Recommended Actions

Risk Area: Overall Governance

Identified factors:

- No material governance threshold was exceeded in this run

Recommended actions:

- Proceed only with controlled monitoring
- Repeat the audit on schedule and after material model changes

Evidence -> Regulation Traceability Matrix

Framework / Requirement	Evidence	Status
EU AI Act Art. 9 Risk Management	Risk score and mitigations generated	PASS
EU AI Act Art. 10 Data Governance / Bias	SPD, DI, EOD fairness evidence	PASS
EU AI Act Art. 13 Transparency	Explainability evidence generated	PASS
EU AI Act Art. 14 Human Oversight	Human-review mitigation included	PARTIAL
EU AI Act Art. 15 Robustness	Noise and missingness stress tests	PASS
UK FCA Principle 6	Group-level outcome monitoring and mitigations	PASS
UK FCA Principle 7	Reason-code style explanation artefacts	PASS
UK PRA Model Risk	Stress testing and governance documentation	PASS

Framework / Requirement	Evidence	Status
NIST AI RMF	Measure and Manage evidence included	PASS
ISO/IEC 42001	Governance and monitoring control suggestions	PARTIAL

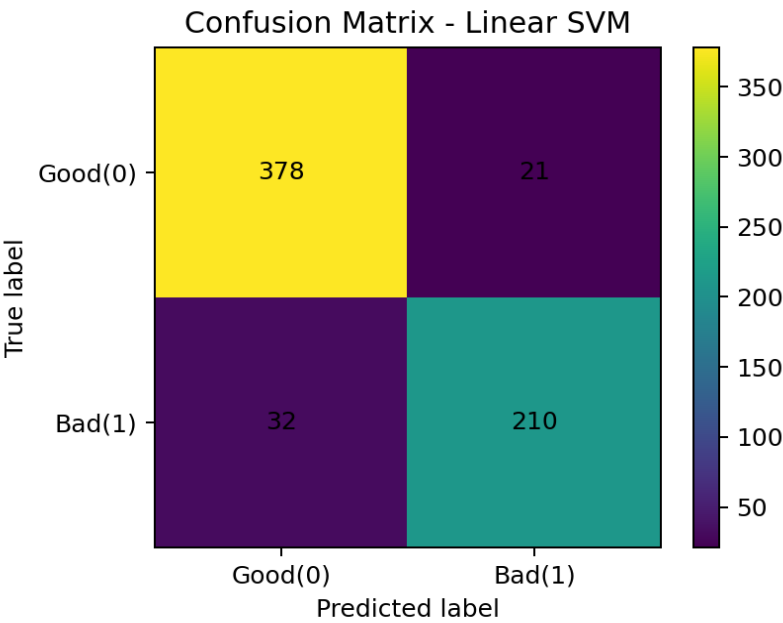
Model Card and Audit - Linear SVM

Performance Score: 90.74
Governance Ready Scoring Engine (GRSE) Score: 28.48 (lower = stronger governance readiness)
GAR Classification: Green
Deployment Decision: Low Risk - Deployable

Predictive Performance

Metric	Validation	Test
Accuracy	0.931	0.917
Precision	0.906	0.909
Recall	0.913	0.868
F1	0.909	0.888
ROC-AUC	0.978	0.967

Confusion Matrix



Governance Evidence

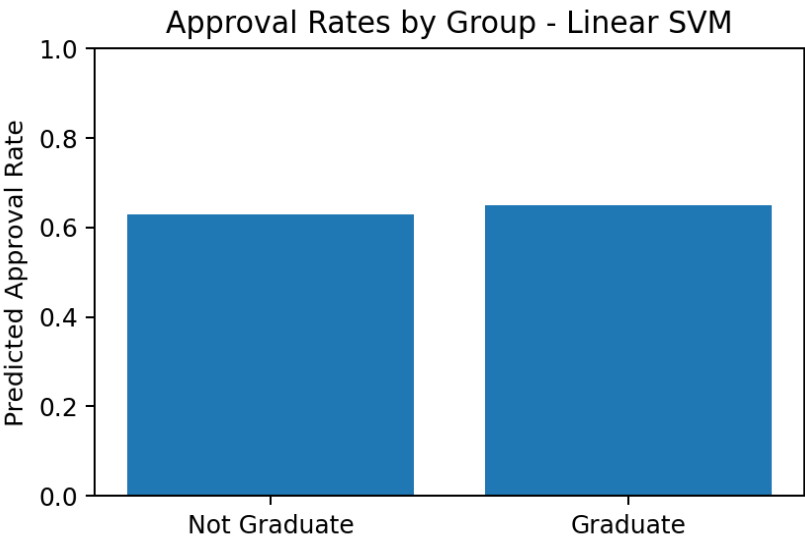
Metric	Value
Approval rate (Not Graduate)	0.629
Approval rate (Graduate)	0.650
SPD: unprivileged - privileged	-0.021
DI: unprivileged / privileged	0.968
EOD: unprivileged - privileged	-0.020
Fairness risk	0.103
Flip-rate (noise)	0.028
Flip-rate (missingness)	0.009
Robustness risk	0.187

Metric	Value
Explainability risk	0.406
Top feature dominance	0.462

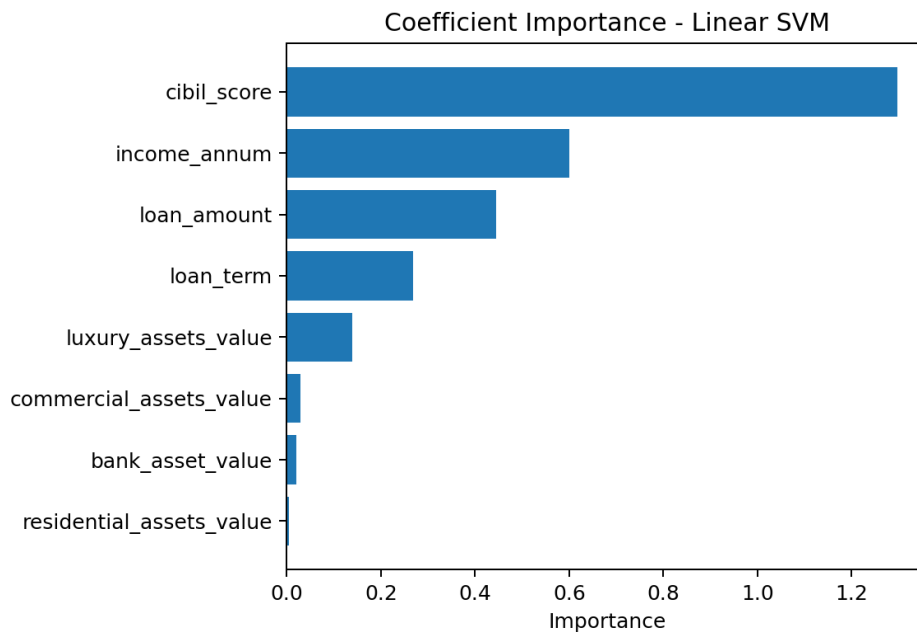
Governance Ready Scoring Engine (GRSE: lower score = stronger governance readiness)

GRSE Component	Risk Score Contribution
Fairness component	3.59
Robustness component	3.74
Explainability component	10.14
Governance baseline component	11.00
Total GRSE score (lower = stronger governance readiness)	28.48

Fairness Evidence



Explainability Evidence



Top Features

Feature	Importance
cibil_score	1.297005
income_annum	0.600466
loan_amount	0.445657
loan_term	0.267721
luxury_assets_value	0.139986
commercial_assets_value	0.029549

Governance Mitigation Factors and Recommended Actions

Risk Area: Transparency and Explainability

Identified factors:

- Top-feature dominance share=0.462
- Possible over-reliance on a small set of drivers

Recommended actions:

- Provide model-level and case-level explanation summaries
- Review dominant features for governance suitability
- Translate technical signals into reason-code style outputs

Evidence -> Regulation Traceability Matrix

Framework / Requirement	Evidence	Status
EU AI Act Art. 9 Risk Management	Risk score and mitigations generated	PASS
EU AI Act Art. 10 Data Governance / Bias	SPD, DI, EOD fairness evidence	PASS
EU AI Act Art. 13 Transparency	Explainability evidence generated	PARTIAL
EU AI Act Art. 14 Human Oversight	Human-review mitigation included	PARTIAL
EU AI Act Art. 15 Robustness	Noise and missingness stress tests	PASS
UK FCA Principle 6	Group-level outcome monitoring and mitigations	PASS
UK FCA Principle 7	Reason-code style explanation artefacts	PARTIAL

Framework / Requirement	Evidence	Status
UK PRA Model Risk	Stress testing and governance documentation	PASS
NIST AI RMF	Measure and Manage evidence included	PASS
ISO/IEC 42001	Governance and monitoring control suggestions	PARTIAL

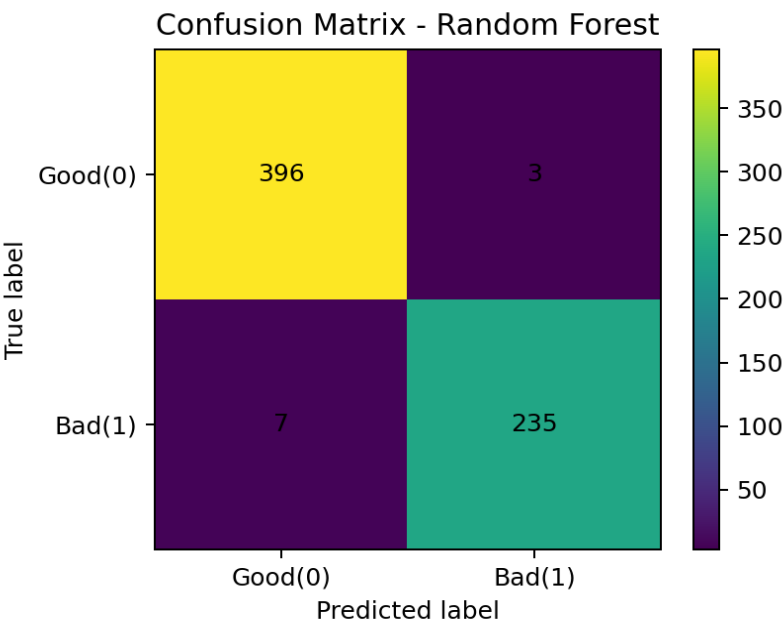
Model Card and Audit - Random Forest

Performance Score: 98.31
Governance Ready Scoring Engine (GRSE) Score: 31.80 (lower = stronger governance readiness)
GAR Classification: Amber
Deployment Decision: Medium Risk - Mitigation Required

Predictive Performance

Metric	Validation	Test
Accuracy	0.988	0.984
Precision	0.992	0.987
Recall	0.975	0.971
F1	0.983	0.979
ROC-AUC	0.999	0.997

Confusion Matrix



Governance Evidence

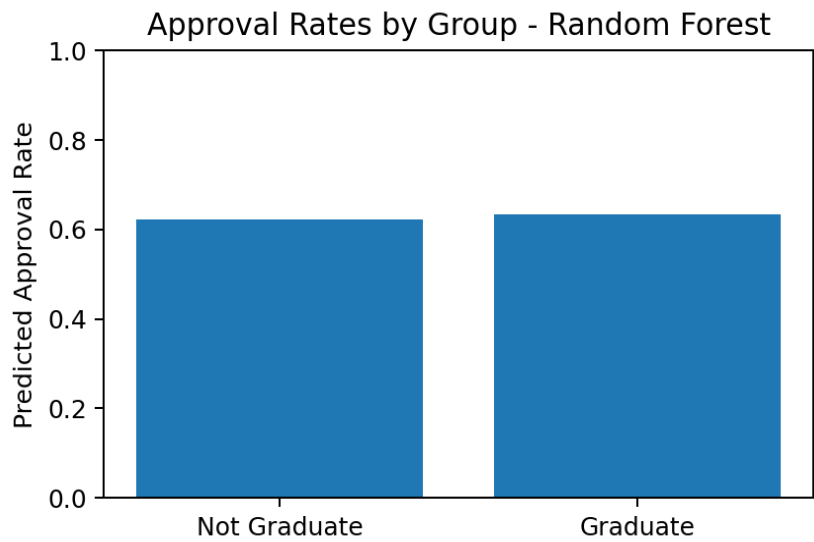
Metric	Value
Approval rate (Not Graduate)	0.623
Approval rate (Graduate)	0.634
SPD: unprivileged - privileged	-0.012
DI: unprivileged / privileged	0.981
EOD: unprivileged - privileged	0.014
Fairness risk	0.072
Flip-rate (noise)	0.006
Flip-rate (missingness)	0.002
Robustness risk	0.039

Metric	Value
Explainability risk	0.700
Top feature dominance	0.797

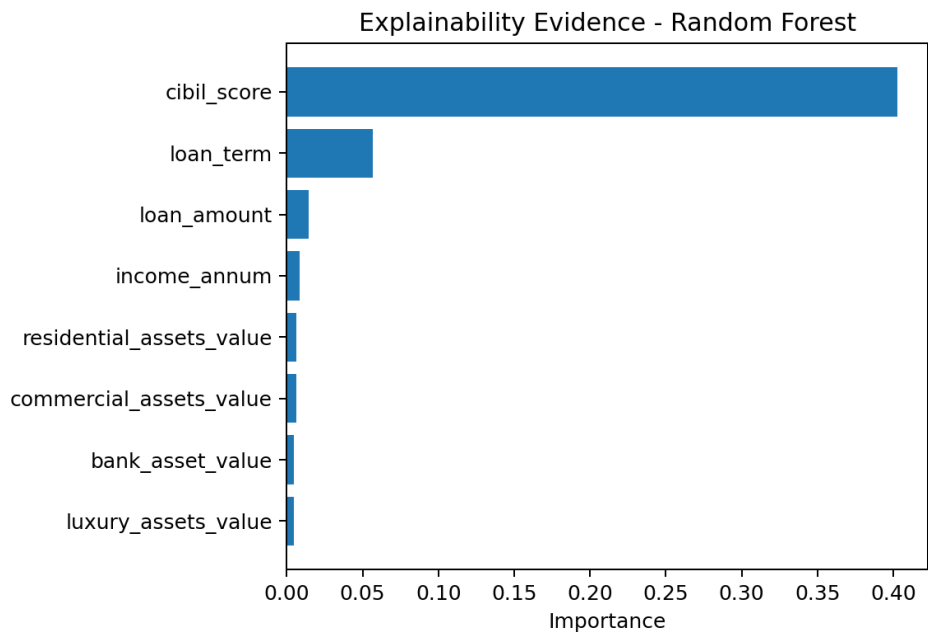
Governance Ready Scoring Engine (GRSE: lower score = stronger governance readiness)

GRSE Component	Risk Score Contribution
Fairness component	2.52
Robustness component	0.78
Explainability component	17.50
Governance baseline component	11.00
Total GRSE score (lower = stronger governance readiness)	31.80

Fairness Evidence



Explainability Evidence



Top Features

Feature	Importance
cibil_score	0.402695
loan_term	0.056723
loan_amount	0.014699
income_annum	0.008409
residential_assets_value	0.006575
commercial_assets_value	0.006420

Governance Mitigation Factors and Recommended Actions

Risk Area: Transparency and Explainability

Identified factors:

- Top-feature dominance share=0.797
- Possible over-reliance on a small set of drivers

Recommended actions:

- Provide model-level and case-level explanation summaries
- Review dominant features for governance suitability
- Translate technical signals into reason-code style outputs

Evidence -> Regulation Traceability Matrix

Framework / Requirement	Evidence	Status
EU AI Act Art. 9 Risk Management	Risk score and mitigations generated	PASS
EU AI Act Art. 10 Data Governance / Bias	SPD, DI, EOD fairness evidence	PASS
EU AI Act Art. 13 Transparency	Explainability evidence generated	FAIL
EU AI Act Art. 14 Human Oversight	Human-review mitigation included	PARTIAL
EU AI Act Art. 15 Robustness	Noise and missingness stress tests	PASS
UK FCA Principle 6	Group-level outcome monitoring and mitigations	PASS
UK FCA Principle 7	Reason-code style explanation artefacts	FAIL

Framework / Requirement	Evidence	Status
UK PRA Model Risk	Stress testing and governance documentation	PASS
NIST AI RMF	Measure and Manage evidence included	PASS
ISO/IEC 42001	Governance and monitoring control suggestions	PARTIAL

Comparative Evaluation - Performance-Centric vs Governance-Centric Ranking

Ranking note: Performance score is ranked highest to lowest. GRSE is a risk score, so governance score is ranked lowest to highest.

Model	Performance Score	Performance Rank	GRSE Score (lower = stronger)	Governance Rank	Ranking Shift
Random Forest	98.31	1	31.80	3	-2
Linear SVM	90.74	2	28.48	2	0
Logistic Regression	90.19	3	28.17	1	2

Mitigation Report - Why Loans Were Not Approved (Dataset-Driven)

Logistic Regression: Common Rejection Drivers

Feature	Effect (Rejected vs Approved)	Direction	Human-readable reason
cibil_score	-1.686	lower	Low credit score (CIBIL)
loan_term	0.235	higher	Unfavourable or long loan term
residential_assets_value	-0.154	lower	Low residential asset value
loan_amount	-0.091	lower	High requested loan amount
income_annum	-0.066	lower	Lower income relative to requested loan
luxury_assets_value	-0.062	lower	Low total asset strength

Example applicant-facing reasons (sample rejected cases)

- Case #0: Low credit score (CIBIL); Unfavourable or long loan term; Lower income relative to requested loan
- Case #7: Low credit score (CIBIL); Lower income relative to requested loan; Low residential asset value
- Case #10: Low credit score (CIBIL); Lower income relative to requested loan; Low bank assets or liquidity
- Case #11: Low credit score (CIBIL); High requested loan amount; Low total asset strength
- Case #14: Low credit score (CIBIL); Unfavourable or long loan term; Low total asset strength
- Case #17: Low credit score (CIBIL); High requested loan amount; Unfavourable or long loan term
- Case #21: Low credit score (CIBIL); Lower income relative to requested loan
- Case #24: Low credit score (CIBIL); High requested loan amount; Unfavourable or long loan term

Suggested applicant mitigation steps

- Improve credit score by paying dues on time and reducing utilisation.
- Adjust loan term to improve affordability and risk profile.
- Provide stronger collateral or better asset evidence.
- Reduce the requested loan amount or strengthen collateral.
- Increase verifiable income or reduce requested exposure.

Linear SVM: Common Rejection Drivers

Feature	Effect (Rejected vs Approved)	Direction	Human-readable reason
cibil_score	-1.681	lower	Low credit score (CIBIL)
loan_term	0.219	higher	Unfavourable or long loan term
residential_assets_value	-0.133	lower	Low residential asset value
loan_amount	-0.079	lower	High requested loan amount
income_annum	-0.035	lower	Lower income relative to requested loan
bank_asset_value	-0.033	lower	Low bank assets or liquidity

Example applicant-facing reasons (sample rejected cases)

- Case #0: Low credit score (CIBIL); Unfavourable or long loan term; Lower income relative to requested loan
- Case #7: Low credit score (CIBIL); Lower income relative to requested loan
- Case #10: Low credit score (CIBIL); Lower income relative to requested loan; Low bank assets or liquidity
- Case #11: Low credit score (CIBIL); High requested loan amount; Low total asset strength

- Case #14: Low credit score (CIBIL); Unfavourable or long loan term; Low total asset strength
- Case #17: Low credit score (CIBIL); High requested loan amount; Unfavourable or long loan term
- Case #21: Low credit score (CIBIL); Lower income relative to requested loan; Low residential asset value
- Case #24: Low credit score (CIBIL); High requested loan amount; Unfavourable or long loan term

Suggested applicant mitigation steps

- Improve credit score by paying dues on time and reducing utilisation.
- Adjust loan term to improve affordability and risk profile.
- Provide stronger collateral or better asset evidence.
- Reduce the requested loan amount or strengthen collateral.
- Increase verifiable income or reduce requested exposure.

Random Forest: Common Rejection Drivers

Feature	Effect (Rejected vs Approved)	Direction	Human-readable reason
cibil_score	-1.643	lower	Low credit score (CIBIL)
loan_term	0.173	higher	Unfavourable or long loan term
residential_assets_value	-0.095	lower	Low residential asset value
luxury_assets_value	0.071	higher	Low total asset strength
income_annum	0.025	higher	Lower income relative to requested loan
loan_amount	0.022	higher	High requested loan amount

Example applicant-facing reasons (sample rejected cases)

- Case #7: Low credit score (CIBIL) (lower; value=465); Unfavourable or long loan term (higher; value=8); Low residential asset value (lower; value=11700000)
- Case #10: Low credit score (CIBIL) (lower; value=477); Unfavourable or long loan term (higher; value=6); Low residential asset value (lower; value=4500000)
- Case #11: Low credit score (CIBIL) (lower; value=478); Unfavourable or long loan term (higher; value=4); Low residential asset value (lower; value=4000000)
- Case #14: Low credit score (CIBIL) (lower; value=397); Unfavourable or long loan term (higher; value=20); Low residential asset value (lower; value=6700000)
- Case #16: Low credit score (CIBIL) (lower; value=543); Unfavourable or long loan term (higher; value=12); Low residential asset value (lower; value=21900000)
- Case #17: Low credit score (CIBIL) (lower; value=329); Unfavourable or long loan term (higher; value=18); Low residential asset value (lower; value=5600000)
- Case #23: Low credit score (CIBIL) (lower; value=509); Unfavourable or long loan term (higher; value=10); Low residential asset value (lower; value=20700000)
- Case #24: Low credit score (CIBIL) (lower; value=376); Unfavourable or long loan term (higher; value=14); Low residential asset value (lower; value=9800000)

Suggested applicant mitigation steps

- Improve credit score by paying dues on time and reducing utilisation.
- Adjust loan term to improve affordability and risk profile.
- Provide stronger collateral or better asset evidence.
- Improve overall asset support or reduce exposure.
- Increase verifiable income or reduce requested exposure.